

ReplicationPad2d#

<https://pytorch.org/docs/stable/generated/torch.nn.ReplicationPad2d.html>

Description:

Pads the input tensor using replication of the input boundary. For N-dimensional padding, use `torch.nn.functional.pad()`. padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 4-tuple, uses (`padding_left\text{padding_left}``padding_left`, `padding_right\text{padding\_right}``padding_right`, `padding_top\text{padding_top}``padding_top`, `padding_bottom\text{padding\_bottom}``padding_bottom`) Note that the output dimensions must remain positive. Input: (N, C, H_{in}, W_{in}) or (N, C, H_{in}, W_{in}) or (C, H_{in}, W_{in}) or (C, H_{in}, W_{in}) . Output: (N, C, H_{out}, W_{out}) or (N, C, H_{out}, W_{out}) or (C, H_{out}, W_{out}) or (C, H_{out}, W_{out}) , where

Function Signature

```
class torch.nn.ReplicationPad2d(padding)[source]#
```

Parameters

Shape:

Code Examples

```
>>> m = nn.ReplicationPad2d(2)
>>> input = torch.arange(9, dtype=torch.float).reshape(1, 1, 3,
3)
>>> input
tensor([[[[0., 1., 2.],
          [3., 4., 5.],
          [6., 7., 8.]]]]])
>>> m(input)
tensor([[[[0., 0., 0., 1., 2., 2., 2.],
          [0., 0., 0., 1., 2., 2., 2.],
          [0., 0., 0., 1., 2., 2., 2.],
          [3., 3., 3., 4., 5., 5., 5.],
          [6., 6., 6., 7., 8., 8., 8.],
          [6., 6., 6., 7., 8., 8., 8.],
          [6., 6., 6., 7., 8., 8., 8.]]]]])
>>> # using different paddings for different sides
>>> m = nn.ReplicationPad2d((1, 1, 2, 0))
>>> m(input)
tensor([[[[0., 0., 1., 2., 2.],
          [0., 0., 1., 2., 2.],
          [0., 0., 1., 2., 2.],
          [3., 3., 4., 5., 5.],
          [6., 6., 7., 8., 8.]]]]])
```

Detailed Documentation

Documentation

Pads the input tensor using replication of the input boundary.

For N-dimensional padding, use `torch.nn.functional.pad()`.

`padding` (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 4-tuple, uses (`padding_left` $\text{\texttt{\text{padding_left}}}$ `padding_left`, `padding_right` $\text{\texttt{\text{padding_right}}}$ `padding_right`, `padding_top` $\text{\texttt{\text{padding_top}}}$ `padding_top`, `padding_bottom` $\text{\texttt{\text{padding_bottom}}}$ `padding_bottom`) Note that the output dimensions must remain positive.

Input: (N,C,Hin,Win)(N, C, H_{in} , W_{in})(N,C,Hin,Win) or (C,Hin,Win)(C, H_{in} , W_{in})(C,Hin,Win).

Output: (N,C,Hout,Wout)(N, C, H_{out} , W_{out})(N,C,Hout,Wout) or (C,Hout,Wout)(C, H_{out} , W_{out})(C,Hout,Wout), where

$H_{out} = H_{in} + \text{\texttt{\text{padding_top}}} + \text{\texttt{\text{padding_bottom}}}$
 $H_{out} = H_{in} + \text{\texttt{\text{padding_top}}} + \text{\texttt{\text{padding_bottom}}}$

$W_{out} = W_{in} + \text{\texttt{\text{padding_left}}} + \text{\texttt{\text{padding_right}}}$
 $W_{out} = W_{in} + \text{\texttt{\text{padding_left}}} + \text{\texttt{\text{padding_right}}}$